

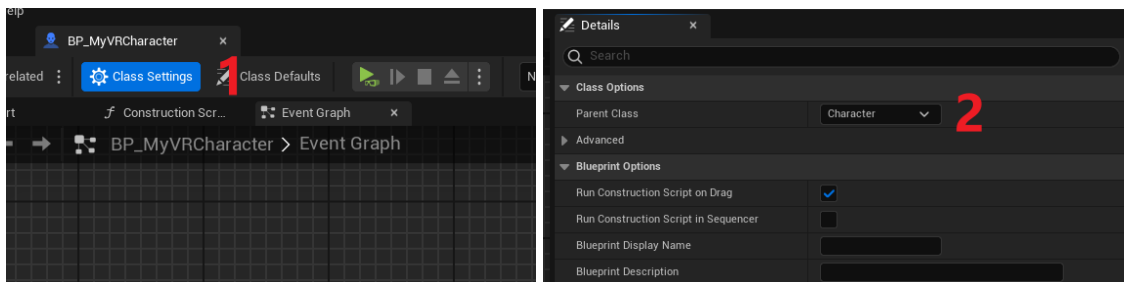
VRPawn Template Modification Steps

Step 1: Change the VRPawn Parent Class to Character

Purpose: Change the VRPawn parent class from Pawn to Character so that it gains the support of the CharacterMovementComponent, including gravity, movement, and navigation.

How to do it:

- Open the VRPawn Blueprint → click Class Settings → change the Parent Class from Pawn to Character.



Notes:

- After changing the parent class, the CharacterMovementComponent takes over movement and rotation control. This must be combined with the settings in Step 3, otherwise rotation will behave incorrectly.

Step 2: Adjust Mesh Collision and the VROrigin Position

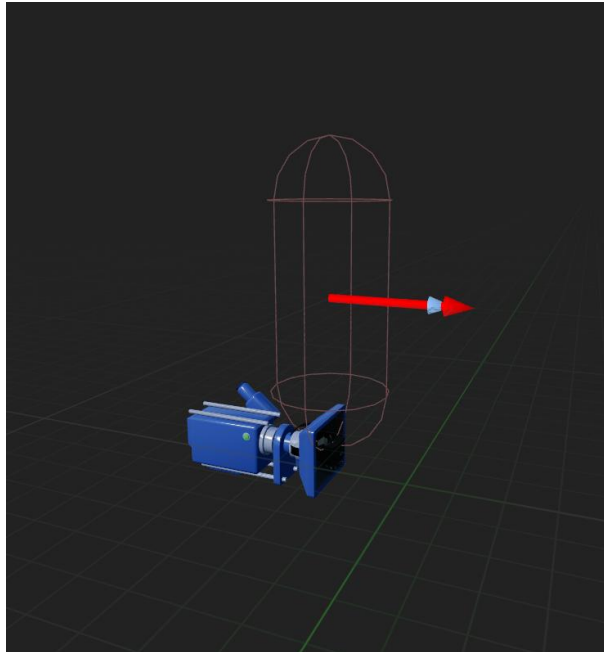
Purpose: Prevent the character's skeletal mesh from physically colliding with scene objects or grabbed objects, and adjust the VROrigin height so the character's feet align with the bottom of the capsule.

2.1 Set Mesh Collision to No Collision

- In the VRPawn Blueprint's Components panel, select the Mesh (Skeletal Mesh) component;
- Find the Collision category in the Details panel;
- Set Collision Presets to No Collision;
- This way the character's body no longer physically collides with anything, avoiding issues such as the character being launched away when grabbing objects.

2.2 Adjust the VROrigin Position

- Select VROrigin (the Scene component);
- In the Details panel, set Transform → Location to (0, 0, -90);
- Lowering VROrigin to the bottom of the collision volume makes the VR camera height measured from the bottom of the capsule, keeping it aligned with the character's collision volume.

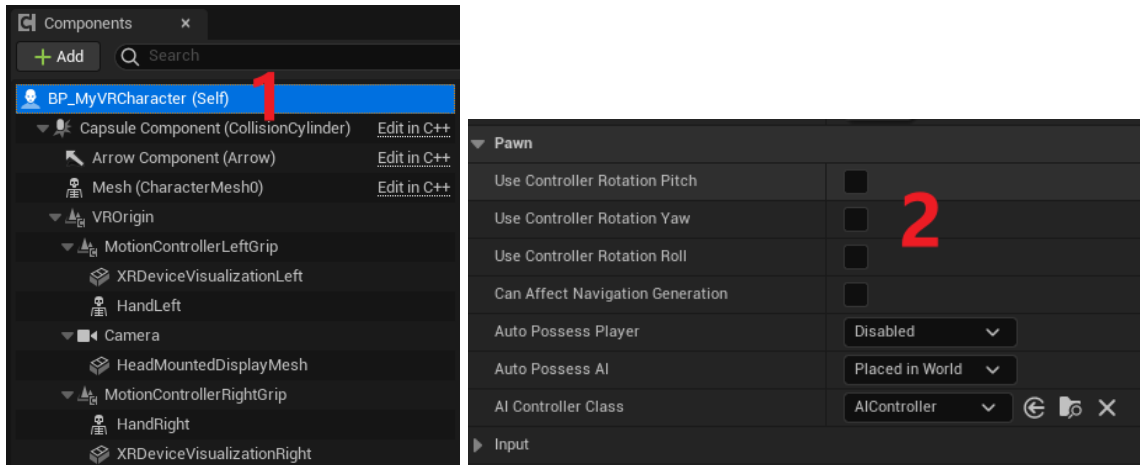


Step 3: Disable Use Controller Rotation Yaw

Purpose: Character enables Use Controller Rotation Yaw by default, which makes the `CharacterMovementComponent` force the capsule to follow the controller's yaw rotation every frame, overriding the VR manual rotation logic and breaking rotation.

How to do it:

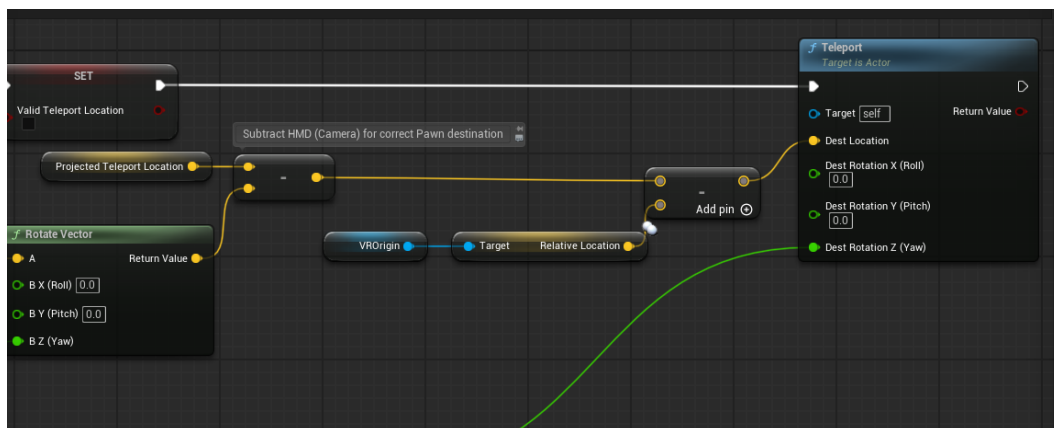
- In the VRPawn Blueprint, click Class Defaults (or select the root component);
- Search for Rotation in the Details panel;
- Uncheck the following options:
 - Use Controller Rotation Yaw
 - Use Controller Rotation Pitch
 - Use Controller Rotation Roll



Step 4: Add HMD Offset Compensation Nodes to the Rotation and Teleport Functions

Purpose: In VR, the player's head (HMD/camera) position is offset from the center of the character capsule. If this offset is not compensated during teleportation, the camera will drift away from the target after teleporting, making the destination feel inaccurate. These nodes calculate and subtract the HMD offset so the camera lands exactly on the target position after teleportation.

Blueprint node screenshots:



Explanation: In VR the player's visual center is the camera (HMD), while teleportation moves the character's root component. If the HMD offset relative to the root component is not subtracted, after teleporting the root component will be at the target position but the camera will be off target. After subtracting VROrigin's relative location, the camera arrives exactly at the target position instead of the root component.

Step 5: Clean Up the Level Environment Actors

Purpose: Remove the VR template's environment actors (showcase models, floors, decorations, etc.) to make room for a custom scene while keeping the basic lighting.

How to do it:

- Open the current level;
- Delete all environment actors in the level (sky atmosphere, volumetric clouds, sky background, fog, etc.);
- Keep only the following two light sources:
 - Directional Light (direct light that simulates the sun)
 - Sky Light (sky light that provides ambient illumination)

Step 6: Add BP_Sky_Sphere

Purpose: Add the sky sphere to give the level a sky background visual. BP_Sky_Sphere automatically adjusts the sky appearance based on the Directional Light's direction and color.

How to do it:

- Search for BP_Sky_Sphere in the Content Browser;
- Drag it into the level;

- Make sure a Directional Light exists in the level; BP_Sky_Sphere will automatically link to it and follow its direction.

Step 7: Set the Sky Light to Static

Purpose: Set the Sky Light's mobility to Static so it participates in static lighting bakes. This reduces runtime computation cost and improves VR performance.

How to do it:

- Select the Sky Light in the level;
- Find Transform → Mobility in the Details panel;
- Change Mobility from Movable to Static.

Notes:

- After switching to Static, the Sky Light no longer updates at runtime. If the sky lighting needs to change dynamically (e.g. day-night cycles), keep it as Movable or Stationary instead.
- For VR scenes, lighting usually does not need to change dynamically, so using Static delivers better performance.